

Runtime-path review (2026-09-07/08)

Four independent reviewers (paper fidelity, FSM logic, integration hazards, Coder/types) audited the deployed path `spoofer_monitor_2hz -> SPF_protectedNav -> SPF_monitorPool -> {SPF_cpiMonitor, SPF_ssMonitor}, SPF_insCoast, SPF_revalidation, STRUCT_SPF, CST_spfParam`. Every finding was then challenged by two independent skeptics; 47 verdicts came back (43 confirmed, 4 refuted, all refuted ones were "info"), 33 verdicts were lost to a session usage limit and are marked *unverified* below.

Confirmed correct (paper lens)

- SPF_cpiMonitor implements Eq. 17/20/29/33/35; per-epoch normalisation makes the H_0 statistic exactly $\text{Gamma}(N/2, 2)$.
- $\Omega = \sigma^2_{\gamma_u} * \sigma^2_t$ is the paper's Eq. 37; Eq. 38 via $\text{gammainc}(T_N/(2(1+\Omega)), N/2)$ reproduces the reference P_{MD} values.
- Baked constants match the offline design to $1e-12$; the per-epoch false-alert budget totals $\leq P_{FA+} + P_{FA-} = 1e-5$.
- SPF_ssMonitor implements Eq. 49-52 with correct time alignment (post-update KF at k vs coast propagated $k-1 \rightarrow k$); window semantics match Fig. 9.
- revalidation's $r'(H P_C H' + R)^{-1} r$ is a valid chi-square(m) test.

Confirmed findings and status

#	Finding	Sev	Status
1	Measurement count frozen at init; any satellite rise/set breaks the pool, CPI and re-validation	blocker	fixed (MAX_MEAS buffers + per-epoch numMeas; <code>tests/test_variable_numMeas.m</code>)
2	Re-validation gate is a single $\chi^2(16)$ threshold; must be $\chi^2(m)$ per satellite count	major	fixed (REVAL_THRESHOLD_TABLE)
3	Re-validation assumes linear $z = Hx$; host $h(x)$ is nonlinear	major	fixed (residual = innovation - H (coast - prior))
4	Anchor bring-forward replays every past interval with the current Φ_{acc}/Q_{acc}	major	fixed (anchor kept live, propagated every epoch)
5	$\sigma_{SS} = 0$ turns the SS gate into "alarm on any nonzero separation"	minor	fixed (undefined test -> no alarm)
6	alarmPerAxis indexed by state index in SPF_monitorPool but by bank position in SPF_ssMonitor	major	fixed (bank position everywhere)
7	Struct fields change class at runtime (info.mode, anchor.epoch, eventAnchorEpoch, freeSlot) — Coder rejects	major	fixed
8	anchorMissing branch (alarm before the first clean close) freezes on the spoofed posterior	major	mitigated : during the warm-up the anchor follows the solution, so the first armed epoch has a fresh anchor; flag still only reported on the latch epoch
9	MAX_ANCHOR_AGE can never trigger at a latch (anchor is 1 epoch old); the drift budget it was meant to enforce is a coast-time budget	major	documented ; coast-time budget deferred by the user; <code>info.coastEpochs</code> telemetry added
10	No maximum coast time / dead-reckoning declaration (guide rule 4)	major	deferred by the user until the system is integrated and tested

#	Finding	Sev	Status
11	run_recovery tests <code>isnan(t_anchor)</code> but the sentinel is 0	minor	fixed
12	Coast = <code>Phi_acc * state</code> is the error-state time update, not an INS mechanization; meaning of <code>kfState / reset_state</code> depends on the host's feedback convention	blocker	fixed: engine rewritten in update-increment form (<code>x+ - x_bar</code>), host owns its filters, gate returns corrections applied through the host's own <code>setKF</code> (see <code>docs/HOST_2HZ_WIRING.m</code>)
13	<code>kfState</code> mirror + Eq. 49 collapse if the host zeroes its error state after feedback	blocker	fixed: the host reduces <code>KF.states</code> by <code>gain*states</code> after feedback (case A); separation is now the accumulated increment <code>d <- Phi d + (x+ - x_bar)</code> , invariant to that bookkeeping
14	<code>reset_state / reseed / feedback</code> semantics only defined for the linear total-state harness	major	fixed: <code>nav.applyCorrection/correction/covar</code> at latch and commit, <code>kfCommand.startTrial = copy the coasting KF</code> ; harness (<code>recovery_nav_sim, scheduler_sim</code>) follows the same contract
15	PROBATION trusts the host to apply the reseed and never verifies it	minor	documented (contract in <code>spoofer_monitor_2hz</code> header)
16	Host KF left free-running on spoofed GNSS after LATCH/VETO until the next probation reseed	minor	by design (fed architecture); documented
17	Persistent sys/epoch: a host <code>clear functions</code> silently re-arms feedback	major	documented; host must pass <code>resetRequest</code> deliberately
18	SS window runs N-1 tests / coast propagations while CPI uses N innovations	info	by design (CPI-aligned window); documented
19	Anchor is the clean window's INS-only coast rather than the window's final KF solution	info	by design (conservative; the coast never ingested the suspect GNSS)
20	<code>altXCheck</code> struct unused; no independent altitude source	info	left for a later baro cross-check

Unverified (skeptical pass lost to the usage limit)

- PROBATION/handback require the host EKF to run open-loop for 18 epochs and accept a reseed (integration).
- Eq. 50 variance can go negative/~0 if the host's own P propagation differs from `spooferInfo.phiAcc/qAcc` (numerical) — mitigated by #5.
- CPI assumes a joint innovation vector with the full S against the prior (sequential-scalar hosts would need a change).
- NaN/Inf from an ill-posed host S or P are swallowed silently.
- Pool struct (~1.3 MB) copied by value several times per epoch; `obsMatrixBuffer` stores all 60 columns although only the position columns are used.
- "Deterministic rounding" scalar loops in `SPF_cpiMonitor` follow two `mldivide` calls, so they do not buy bit-exactness.
- Coast covariance symmetrised in `SPF_insCoast` but not in `SPF_ssMonitor` (no observable effect).